Projector-Equivalent Wilson–Dirac Hamiltonians and Dynamical Identification

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Abstract. A strictly positive finite-range scalar deformation of a gapped two-band Wilson–Dirac Hamiltonian leaves the negative-energy band projector, hence the zero-temperature gauge-invariant quasifree state, unchanged while altering the excitation spectrum. We separate that exact state-level degeneracy from the ideal dynamical measurements that distinguish the Hamiltonians. We provide an algebraic three-momentum necessary relation, an arbitrary-six-node degree-restricted harmonic discriminator, and a conditional exact mass-and-deformation reconstruction. Each mathematical assertion has an explicit mass branch, observation class, and boundary condition. The counterexamples concern negative-mass reciprocal branches, higher-harmonic interpolation aliases, uncertain momenta, and finite open-chain truncation.

Keywords: Wilson–Dirac operators; occupied-band projectors; inverse spectral problems; finite-range deformations.

1. Model and theorem domains

The quasifree-state conventions use the canonical anticommutation relations as developed by Araki [1]. The finite-node identifiability arguments use standard polynomial interpolation [2].

Fix the infinite, translation-invariant one-dimensional two-orbital chain and the common CAR algebra. With $k \in [-\pi, \pi]$ and $m > 0$, define

$$h_m(k) = \sin k\, \sigma_x + (m + 1 - \cos k)\sigma_z, \qquad E_m(k) = \sqrt{m^2 + 4(m+1)\sin^2(k/2)}.$$

Let $g_\epsilon(k) = 1 + \epsilon \cos k$ with $|\epsilon| < 1$, and $h_{m,\epsilon}(k) = g_\epsilon(k) h_m(k)$. The one-particle spectral gap at momentum k is $2g_\epsilon(k)E_m(k)$. Here 'gap' means *interband splitting at a momentum*, not a continuum quantum-field-theoretic mass gap. At $m = 0$ the isolated zero-energy fiber is irrelevant to the infinite-chain covariance almost everywhere; a finite discrete zero mode requires an occupation convention.

Theorem 1 (exact zero-temperature projector equivalence). For every $m > 0$ and $|\epsilon| < 1$,

$$P_{m,\epsilon}(k) = P_m(k) = \frac{1}{2}\left[I - \frac{h_m(k)}{E_m(k)}\right].$$

Proof. Since $h_m(k)^2 = E_m(k)^2 I$, multiplication by the strictly positive scalar $g_\epsilon(k)$ changes eigenvalue magnitudes but not eigenspaces or signs. Its Fourier symbol is trigonometric polynomial of degree at most two, so the deformed hopping remains finite range. The two occupied-band projectors coincide pointwise. A gauge-invariant quasifree state is determined by its one-particle covariance, proving identity of every finite-interval restriction and equal-time correlation on a *common, fixed algebra.* □

This theorem does not assert equality of energy measurements whose measuring operator changes with the Hamiltonian, of positive-temperature Gibbs states, or of finite *open-boundary* Hamiltonian projectors. Compression of a full-line spectral projector is not the spectral projector of a truncated open-chain Hamiltonian.

2. Ideal dynamics supplies information missing from the projector

For a momentum sector prepared with its lower band occupied and upper band empty, the interband operator $O_k = c^\dagger_{+,k} c_{-,k} + c^\dagger_{-,k} c_{+,k}$ obeys, in the ideal closed model,

$$\langle O_k(t) O_k(0) \rangle = e^{-i\Lambda\,[2g_\epsilon(k) E_m(k)]\,t/\hbar}.$$

A common unknown positive time-to-energy factor $\Lambda/\hbar$ cancels in ratios of correctly identified angular frequencies. This observation class requires coherent interband access, a common clock, calibrated momentum labels, and a chosen band occupation; the mathematical formula alone does not certify that instrument.

Theorem 2 (three-momentum necessary relation). For $m = 1$, $k_1 = \pi/3$, $k_2 = 2\pi/3$, and $k_3 = \pi$, define $R_{12} = \omega(k_1)/\omega(k_2)$ and $R_{32} = \omega(k_3)/\omega(k_2)$. Then

$$R_{12} = \sqrt{3/7}\,\frac{1+\epsilon/2}{1-\epsilon/2}, \qquad R_{32} = \frac{3(1-\epsilon)}{\sqrt{7}(1-\epsilon/2)}, \qquad \boxed{\sqrt{21} R_{12} + 2\sqrt{7} R_{32} = 9}.$$

The first ratio is strictly increasing in ϵ, so it identifies ϵ *inside this fixed-mass family.* The final equality is necessary but not sufficient for membership. For a periodic finite chain the three momenta require chain length divisible by six.

3. Finite-degree harmonic extensions

Put $c = \cos k$, $s = m + 1 > 1$, $\lambda > 0$, and let the strictly positive deformation be $g(c) = 1 + bc + tc^2$. The squared positive thermal splitting is

$$Q(c) = D(c)^2 = \lambda^2 g(c)^2 (s^2 + 1 - 2sc).$$

It is a polynomial of degree at most five with leading coefficient $[c^5]Q = -2\lambda^2 s t^2$.

Theorem 3 (six-node discriminator). At arbitrary six *distinct, exactly known* cosine nodes $c_0, \ldots, c_5$,

$$\boxed{\sum_{i=0}^{5} \frac{Q(c_i)}{\prod_{j \neq i} (c_i - c_j)} = -2\lambda^2 s t^2.}$$

Proof. Lagrange interpolation reconstructs the polynomial Q uniquely and its leading coefficient equals the displayed divided difference. □ It follows that within the specified quadratic-harmonic class the signed statistic vanishes exactly when $t = 0$. For $g(c) = \sum_{j=0}^{r} a_j c^j$, the analogous $(2r+1)$-st divided difference from $2r+2$ distinct nodes is $-2\lambda^2 s a_r^2$. Neither theorem rules out unrestricted higher harmonics.

Theorem 4 (exact six-node injection with a mass-branch condition). With $s > 1$, $g(0) = 1$, $g > 0$ on $[-1, 1]$, and exact $D(c_i) > 0$, the six data determine (m, λ, b, t) uniquely within the normalized quadratic family.

Proof. The interpolated $Q(c)$ has exactly one root of odd multiplicity, $r = (s + s^{-1})/2 > 1$; if the deformation also vanishes there, that root's multiplicity changes by an even amount and remains odd. The condition $s > 1$ fixes $s = r + \sqrt{r^2 - 1}$. The polynomial $Q/(s^2 + 1 - 2sc)$ equals $(\lambda g)^2$, whose strictly positive polynomial square root on the physical interval is fixed; its constant coefficient is λ, and its remaining coefficients give b, t. □

The normalization $g(0) = 1$ is indispensable: without it, a common amplitude can move between g and λ. The branch is indispensable too: $s \mapsto s^{-1}$ and $\lambda \mapsto s\lambda$ leave *all* splittings unchanged, but move m out of the stated positive-mass domain.

4. Statistical identifiability is a separate problem

The signed leading-coefficient witness is quadratic in t, so its derivative with respect to t vanishes at $t = 0$. That does **not** imply that the full six-band occupation data have zero nuisance-profiled local information about t: their first-order variation contains momentum functions not spanned by the first-harmonic nuisance directions. The relevant Fisher statement requires independent finite-temperature occupations, correctly labeled bands, a common offset, and the declared parametric model; it does not extend to arbitrary instruments.

For six cosine nodes, the barycentric formula can be ill conditioned for perturbed momentum labels, near-colliding nodes, or finite-error estimates of Q. A higher-harmonic polynomial that vanishes at all measured nodes produces an exact, indistinguishable finite-data alias. Thus global *exact* parameter injection in a degree-restricted family is not unrestricted function identification.

Counterexample 1 (constructive finite-node higher-harmonic alias; counterexample). Fix a continuous strictly positive deformation $g_0(c)$ on $[-1, 1]$ and finitely many distinct sampled nodes $c_1, \dots, c_M$. Set $H(c) = \prod_{j=1}^{M}(c - c_j)$ and $g_\tau(c) = g_0(c) + \tau H(c)$. For every nonzero τ satisfying $|\tau|\|H\|_\infty < \min_{[-1,1]} g_0$, the deformed symbol remains strictly positive and differs away from the sampled nodes, while $g_\tau(c_j) = g_0(c_j)$ for all j. Hence every sampled splitting and all equal-time occupied-projector observations coincide, although the unsampled spectral functions differ. This countermodel requires allowing polynomials of degree at least M; it does **not** contradict Theorem 3 or Theorem 4 within their declared quadratic-harmonic family. It also does not establish finite-shot indistinguishability under additional off-node measurements.

5. Finite-boundary limitation and exact state-relative-entropy bridge

Finite open-chain counterexample. Multiplying the full-line symbol by g_ϵ *before* truncating to seven open sites generally produces distinct negative-energy projectors. At $m = 1$, $\epsilon = 0.37$, the explicit 14×14 open-chain matrices give $\|P_0^{\text{open}} - P_{0.37}^{\text{open}}\|_{\text{op}} \simeq 0.0065324543$. The specified finite matrices demonstrate that the full-line projector equivalence need not persist for open-chain spectral projections. The open-chain example distinguishes truncating the symbol from projecting the truncated Hamiltonian.

Ground-state relative entropy. On the common infinite-chain CAR algebra and for the same $m > 0$, Theorem 1 gives the *same* gauge-invariant quasifree ground state, so their restrictions to every finite interval have relative entropy **exactly zero**. This statement is about comparing h_m with $g_\epsilon h_m$ at the **same mass**. It is not an assertion that the distinct-mass states in another model have zero relative entropy. At positive inverse temperature β, the one-particle thermal covariance instead depends on the multiplier,

$$C_{\beta,g}(k) = \tfrac{1}{2}\Big[I - \tanh\big(\tfrac{1}{2}\beta g(k)E_m(k)\big)\, h_m(k)/E_m(k)\Big].$$

Consequently calibrated thermal observations can distinguish some deformations invisible to zero-temperature projector observations. There is no implied laboratory calibration or physical selection.

References

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